

ASSESSMENT-CO5-AT2-Pseudocode Writing Task (Algorithm Design)

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SIMATS Online Programming Lab

DAA PROGRAMMING



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1/1 answered

(true/false). Design an algorithm that assigns truth values to variables systematically. Use control structures to evaluate clauses dynamically. Apply pruning to eliminate unsatisfied assignments early. Ensure the pseudocode is structured and readable. Handle all possible cases.

Your Code

```
1 def sat(c, v, a, i):
2     if i == len(v):
3         return all(any(a[abs(x)] == (x > 0) for x in c[i]) for i in range(len(c)))
4
5     a[v[i]] = True
6     if sat(c, v, a, i + 1):
7         return True
8
9     a[v[i]] = False
10    if sat(c, v, a, i + 1):
11        return True
12
13    del a[v[i]]
```

```
14     return False
15
16
17 c = [[1, 2, 3], [-1, 2, 3], [1, -2, 3]]
18 v = [1, 2, 3]
19 a = {}
20
21 print("TRUE" if sat(c, v, a, 0) else "FALSE")
```

Input

stdin input

Output

TRUE

✔ Submitted